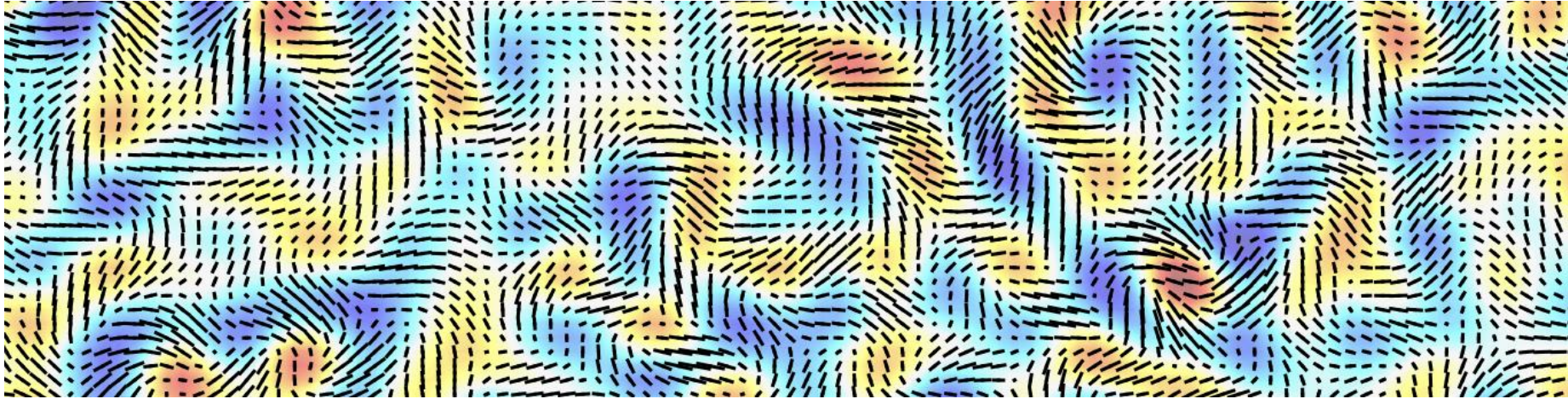




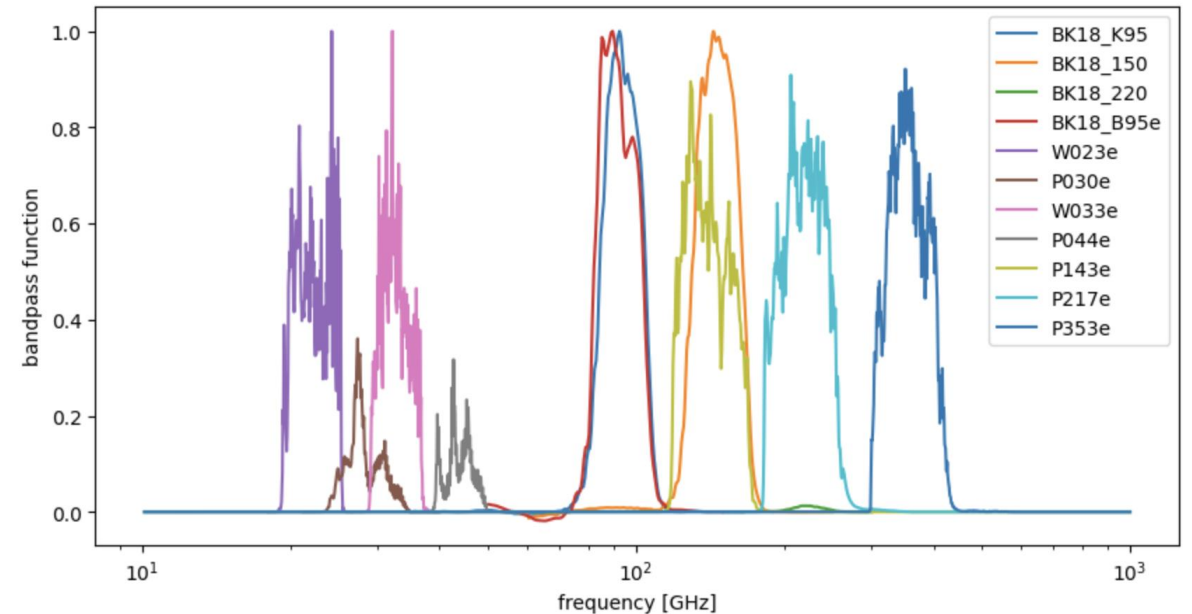
Modeling Galactic Foregrounds for Improved Cosmic Microwave Background (CMB) B-mode Signal

Capstone Project — Arya Papal

Advisor: Dr. Colin Bischoff

Why Cosmic Microwave Background (CMB)?

- The **Cosmic Microwave Background**, or CMB, is relic radiation from the early universe, dating back to about 380,000 years after the Big Bang.
- We're particularly interested in **B-mode polarization** patterns in the CMB.
- Detecting B-modes would provide **direct evidence of primordial gravitational waves**.
- B-mode signal is incredibly faint — only a few **nanokelvins** — and is heavily contaminated by **Galactic foregrounds**.



Instrumental Bandpass Functions centered at different frequencies.

Foregrounds: Dust and Synchrotron

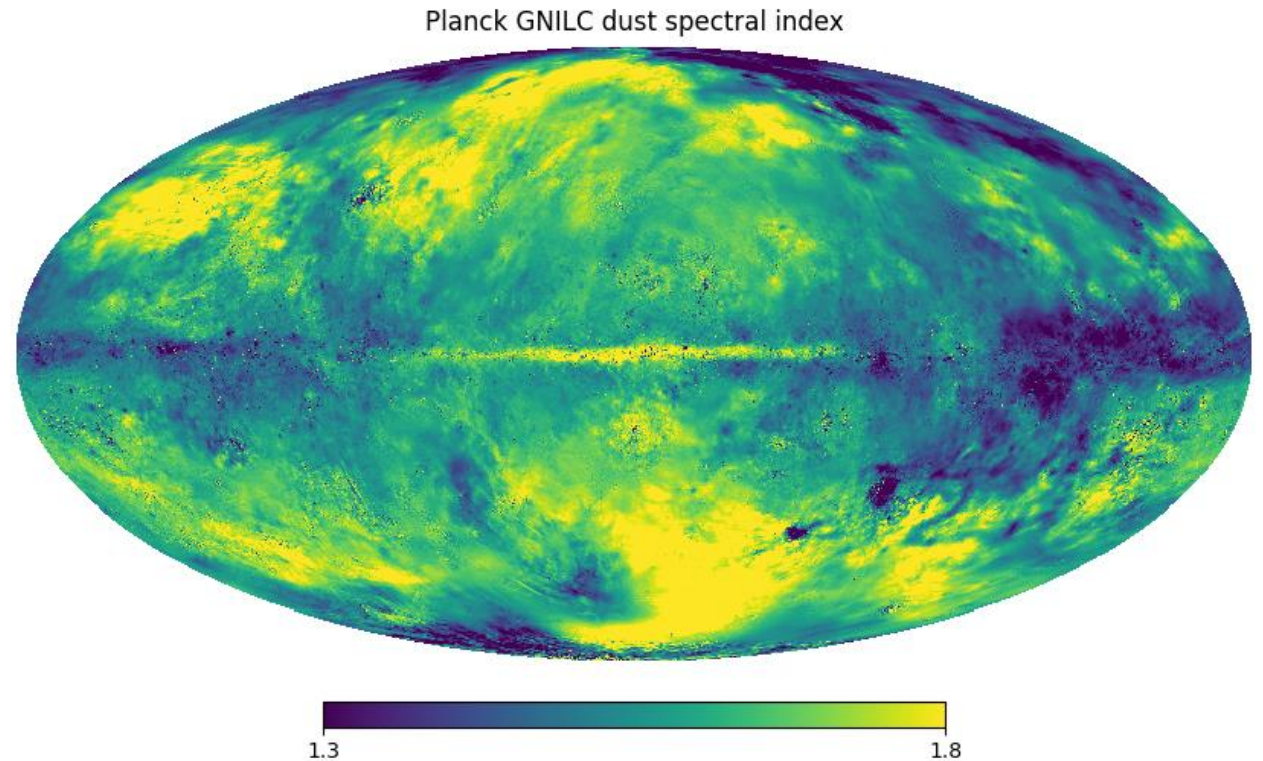
- Foregrounds primarily come from two sources:

-**Synchrotron radiation** at low frequencies

-**Thermal dust emission** at high frequencies.

- While early models assumed uniform dust properties, real observations — from Planck, BICEP/Keck, and others — show **spatial variability** in dust parameters like spectral index β_D .

- This causes **frequency decorrelation**, which can bias B-mode measurements and estimates of r



Simulated map of β_D visualizes the spatial variation across the sky.

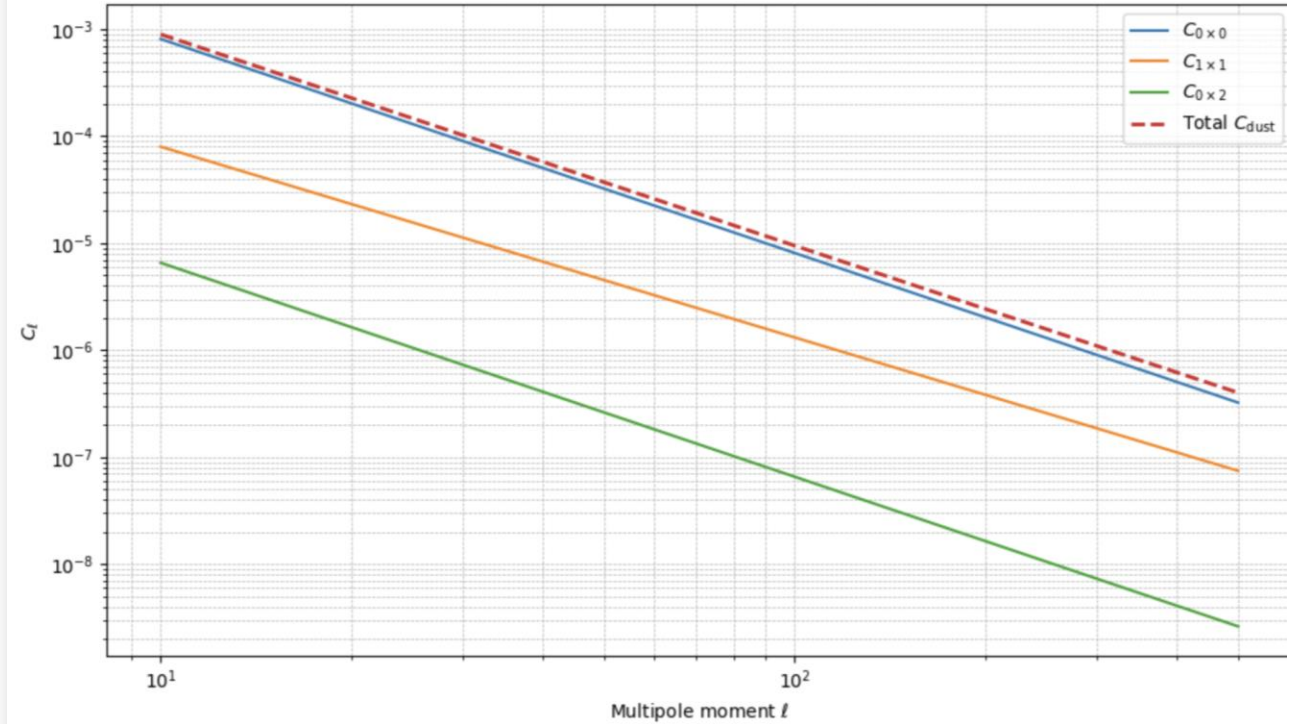
Moments Expansion Method – Theoretical Framework

Instead of assuming constant spectral parameters, this method **Taylor expands** the frequency scaling around a mean value, allowing for spatial fluctuations.

This gives rise to three key terms in the cross-power spectrum between two frequencies ν' and ν expanded as: $C_l^{\nu\nu'} = C_l^{(0\times 0)} + C_l^{(1\times 1)} + C_l^{(0\times 2)}$

- $C_l^{(0\times 0)} = S_\nu(\beta_D) \cdot S_{\nu'}(\beta_D) \cdot C_l^{DD}$, corresponds to the standard cross-spectrum assuming constant β_D
- $C_l^{(0\times 2)} = \sum \frac{1}{2} \cdot (S_\nu(\beta_D) \cdot \partial^2 S_{\nu'}(\beta_D) + \partial^2 S_\nu(\beta_D) \cdot S_{\nu'}(\beta_D)) \cdot C_l^{DD} \sigma^2$, captures variance-driven corrections due to the second-order perturbation
- $C_l^{(1\times 1)} = \sum \partial S_\nu(\beta_D) \cdot \partial S_{\nu'}(\beta_D) \sum_{l_1, l_2} \frac{(2l_1+1)(2l_2+1)}{4\pi} \begin{pmatrix} l & l_1 & l_2 \\ 0 & 0 & 0 \end{pmatrix} C_{l_1}^{DD} \cdot C_{l_2}^\beta$, Correlations of first-order spectral index fluctuations

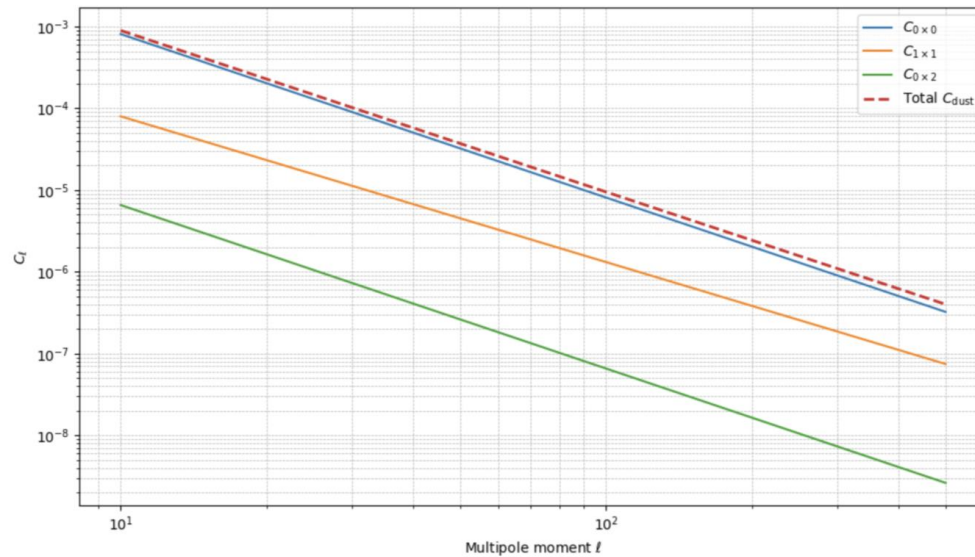
Spectral Energy Distribution (SED): $S_\nu \propto \left(\frac{\nu}{\nu_0}\right)^{\beta_D} \cdot \frac{B_\nu(T_D)}{B_{\nu'}(T_D)}$, here $B_\nu(T_D)$ being the Planck function and ν_0 the pivot frequency



Implementation Pipeline

1. Loaded bandpass data from BICEP/Keck, Planck and WMAP.
2. Calculating Baseline Spectrum $C_l^{0 \times 0}$.
3. Calculating Curvature Term $C_l^{(0 \times 2)}$.
4. Calculating Spectral Index Fluctuation $C_l^{1 \times 1}$.
5. Generated plots showing all 3 moment terms across various frequency combinations.

Understanding the Three Moment Terms



The Three Key Moment Terms:

- $C_l^{(0 \times 0)}$: Baseline spectrum assuming constant dust spectral index β_D

$$C_l^{(0 \times 0)} = S_v(\beta_D) \cdot S_{v'}(\beta_D) \cdot C_l^{DD}$$

- $C_l^{(0 \times 2)}$: Curvature correction accounting for variance in β_D

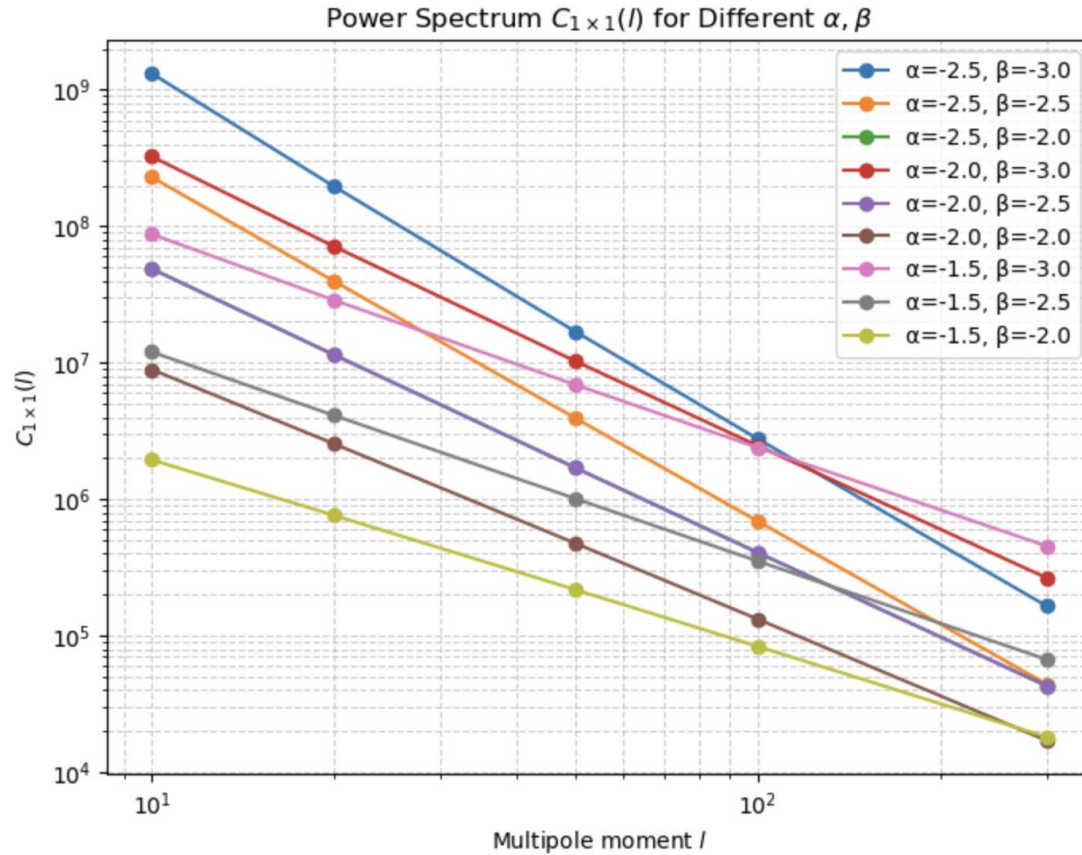
$$C_l^{(0 \times 2)} = \sum \frac{1}{2} \cdot (S_v(\beta_D) \cdot \partial^2 S_{v'}(\beta_D) + \partial^2 S_v(\beta_D) \cdot S_{v'}(\beta_D)) \cdot C_l^{DD} \sigma^2$$

- $C_l^{(1 \times 1)}$: First-order spatial correlation of β_D fluctuations

$$C_l^{(1 \times 1)} = \sum \partial S_v(\beta_D) \cdot \partial S_{v'}(\beta_D) \sum_{l_1, l_2} \frac{(2l_1+1)(2l_2+1)}{4\pi} \begin{pmatrix} l & l_1 & l_2 \\ 0 & 0 & 0 \end{pmatrix} C_{l_1}^{DD} \cdot C_{l_2}^{\beta}$$

These terms sum together to model total dust contamination across frequencies

Power-law Approximation



- The exact computation of $C_l^{1 \times 1}$ was expensive (7-9 hours per run)
- $C_l^{1 \times 1} \approx A_D \left(\frac{l}{l_*}\right)^{\gamma_D}$
 - Regression model used combinations α_β and α_D .
 - γ_D is the effective slope of the moment term.
 - A_D is the amplitude of the moment term at pivot multipole $l^* (80)$.
- This approximation drastically reduces computation time, making the model feasible for integration in MCMC pipelines.

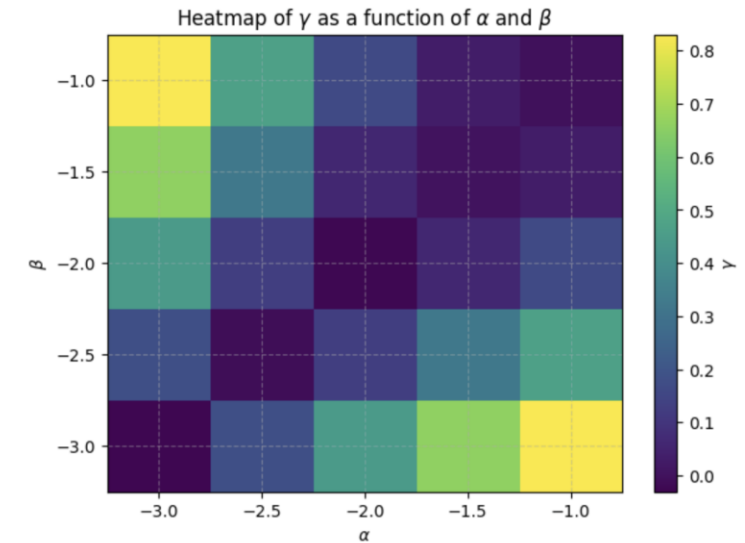
Regression to find A_D and γ_D

Finding the Power-Law Slope γ_D

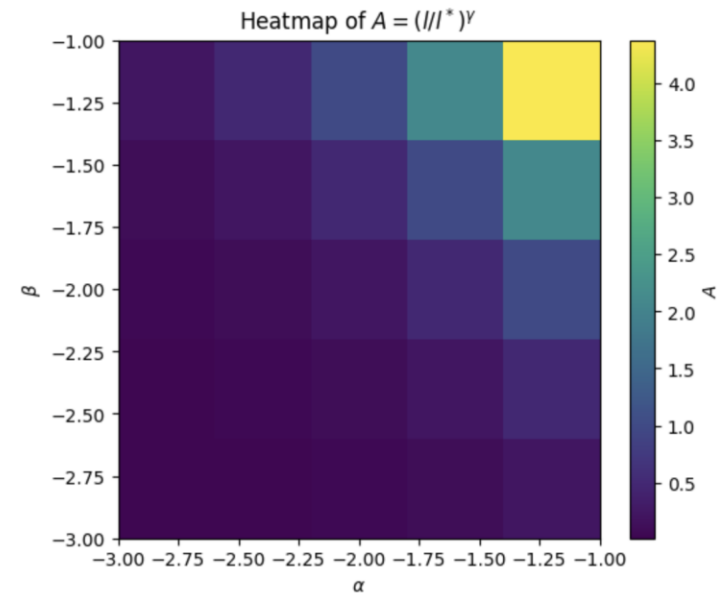
- γ_D is the slope of the 1×1 term in multipole space.
- Performed linear regression on log-log plots of $C_l^{1 \times 1}$ vs ℓ
- Regression used α_D and α_β combinations.

Finding the Power-Law Amplitude A_D

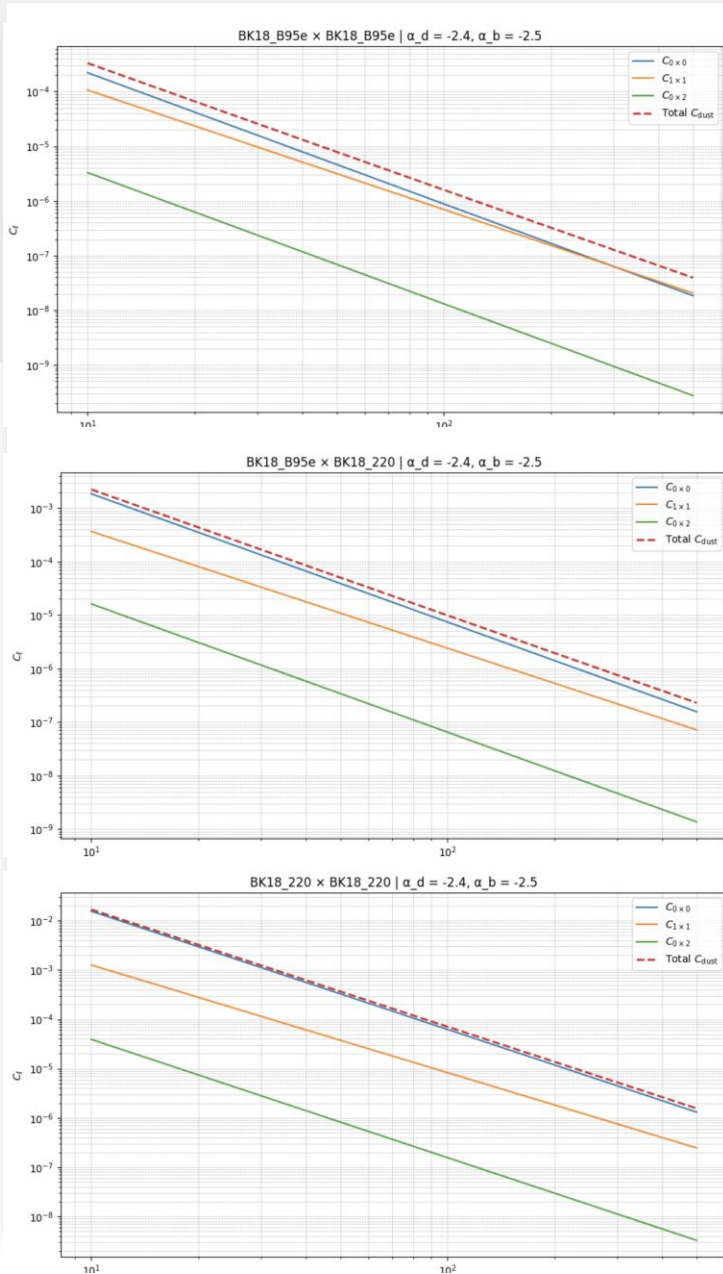
- Once γ_D was known, we calculated A_D using: $A_D = \frac{C_l^{1 \times 1}}{(\frac{l}{l_*})^{\gamma_D}}$
- Captures the scale of the moment term at pivot multipole



Heatmaps of power-law parameters γ_D and A_D as a function of α_D and α_β



Dust Power Spectrum Decomposition



- Each panel shows the decomposition of the dust angular power spectrum into its three main moment expansion components:

- $C_l^{(0 \times 0)}$: Baseline term (blue)
- $C_l^{(1 \times 1)}$: Spectral index variation (orange)
- $C_l^{(0 \times 2)}$: Curvature correction (green)
- $C_l^{vv'} = C_l^{(0 \times 0)} + C_l^{(1 \times 1)} + C_l^{(0 \times 2)}$: Sum of all terms (red dashed)

- Simulated for different frequency channel combinations:

- BK18_B95e x BK18_B95e
- BK18_B95e x BK18_220
- BK18_220 x BK18_220

Future work

- **Extend the framework to include synchrotron emission and its cross-correlations with dust.**
- **Integrate this formalism into full MCMC (Markov Chain Monte Carlo framework) pipelines.**
- **Quantify bias on r in simulated likelihood analyses.**
- **Apply method to real BICEP/Keck and Planck cross-spectra.**

REFERENCES:

- [1] S Azzoni et al. 2021. Phys Rev D 103(8):083517
- [2] M Kamionkowski and E D Kovetz. 2015. Annu Rev Astron Astrophys 54:227–269
- [3] D Samtleben, S Staggs, and B Winstein. 2007. Annu Rev Nucl Part Sci 57:245–283
- [4] W Hu and M White. 1997. New Astron 2(4):323–344
- [5] BICEP/Keck Collaboration. 2018. Phys Rev Lett 121(22):221301